



# *CCO Modernization Roadmap*

*John Beverley*

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Assistant Professor, *University at Buffalo*

Associate Director, *Institute of Artificial Intelligence and Data Science*

# *Outline*

- CCO Governance Board & NCOR
- CCO Modernization Effort
- CCO Future Release Strategy

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# The Common Core Ontologies (CCO)

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## What is CCO?

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The Common Core Ontologies (CCO) is a widely-used suite of eleven ontologies that consist of logically well-defined generic terms and relations among them reflecting entities across all domains of interest.

These eleven ontologies constitute a [mid-level ontology](#) that extends from the [Basic Formal Ontology \(BFO\)](#), an [ISO-standard](#) top-level ontology. Whereas BFO represents only the most generic entities and relations, CCO contains classes that users will find common across data sets in many domains. Such classes include, for example, person, facility, date, employment, nickname, and measurement.

Both BFO and CCO have been directed for use as "[baseline standards](#)" for formal ontology development across the United States Department of Defense and Intelligence Community.

CCO is currently being evaluated as a mid-level ontology standard by the IEEE Standards Association under [PAR3195.1](#).

CCO itself is not intended to grow indefinitely by including content that is proper to particular domains. Users are encouraged to create their own domain extensions with content particular to those domains and publish these ontologies for re-use by others.

<https://github.com/CommonCoreOntology/CommonCoreOntologies>

# Who owns CCO?

CCO is an open-source project. CCO is freely available for reuse under the BSD 3-Clause license. With this said, 'ownership' might be interpreted to mean any of the following:

1. Copyright Holder: As the initial creator and supporter of CCO, CUBRC, Inc is the copyright holder.
2. Licensed User: As the copyright holder, CUBRC, Inc. agreed to license CCO under its copyright so that it can be used with minimal restrictions; specifically, CCO is released under a BSD 3-Clause License. (See: <https://github.com/CommonCoreOntology/CommonCoreOntologies/blob/develop/LICENSING>)
3. Configuration Management: Today, CCO users find CCO on GitHub where it is overseen as an open-source project by the CCO Board of Directors and the CCO Developer Group.

# *CCO Governance Board*

Dear Ontology Community

The Common Core Ontologies (CCO)<sup>1</sup> have become, over the last decade, an increasingly important resource for the U.S. Government. This suite of mid-level ontologies is currently used by dozens of organizations and deployed in critical systems in active operation. To date, CCO has been developed and maintained by Ron Rudnicki and the ontology team at CUBRC, Inc., who have overseen successive releases of CCO that have always sought to meet the needs of a growing number of end users. Through the ingenuity and discipline of this team, CCO has remained a reference mid-level ontology that greatly reduces the time needed to develop jointly interoperable high-quality domain ontologies aligned to Basic Formal Ontology (BFO). As a result, CCO has continued to receive greater adoption, is being proposed as an IEEE standard, and will become a central component of the DoD IC Ontology Foundry.

In recognition of the need for CCO to continue to scale and evolve, in use and in scope, today we announce that CCO will, in future releases, be overseen by a governance board. The Common Core Governance Board will have an established charter with transparent and documented procedures and will be composed of representatives from stake-holder organizations. Initial members will include:

- Mark Jensen (CUBRC, Inc.)
- Alexander Cox (CUBRC, Inc.)
- John Beverley, Ph.D. (The University at Buffalo, SUNY)
- Barry Smith, Ph.D. (The University at Buffalo, SUNY)
- J. Neil Otte, Ph.D. (The Johns Hopkins University Applied Physics Laboratory)

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# ***NCOR 501(c)3***

- Established as a non-profit at the end of 2024
- Board of Directors: Barry Smith & John Beverley
- Accepts donations, service contracts, conference hosting, etc.
- Recently, funded by USAF Air Combat Command to mature CCO into a stable artifact





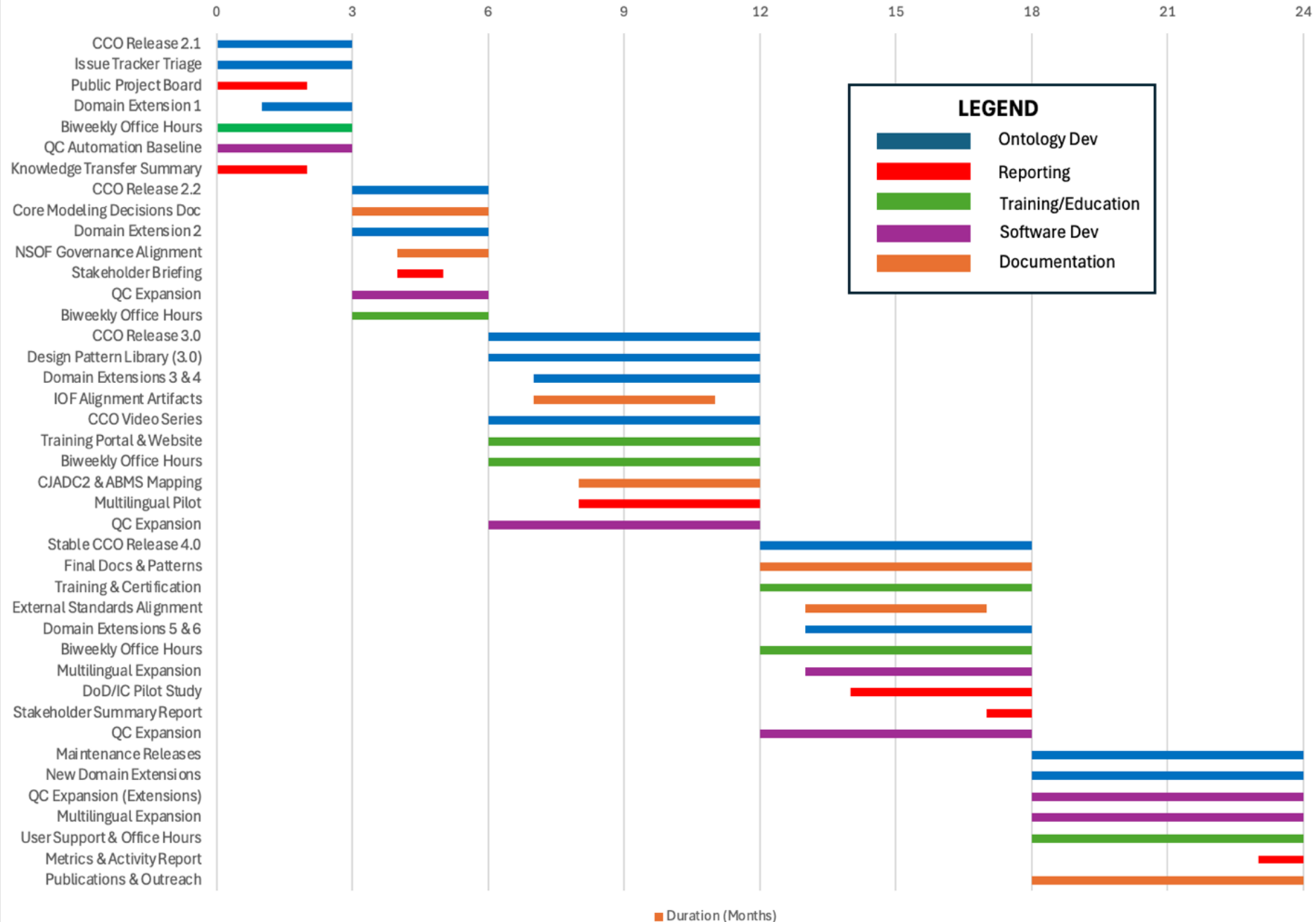
# *CCO Governance Board & NCOR*

- Overlapping membership (Barry & John); John leading CCO modernization effort for NCOR
- As with any changes to CCO, updates ultimately go through governance board members
- For 2.1 release, NCOR coordinated updates to repositories and artifacts
- NCOR to coordinate directly with CCO Developer Group for later releases

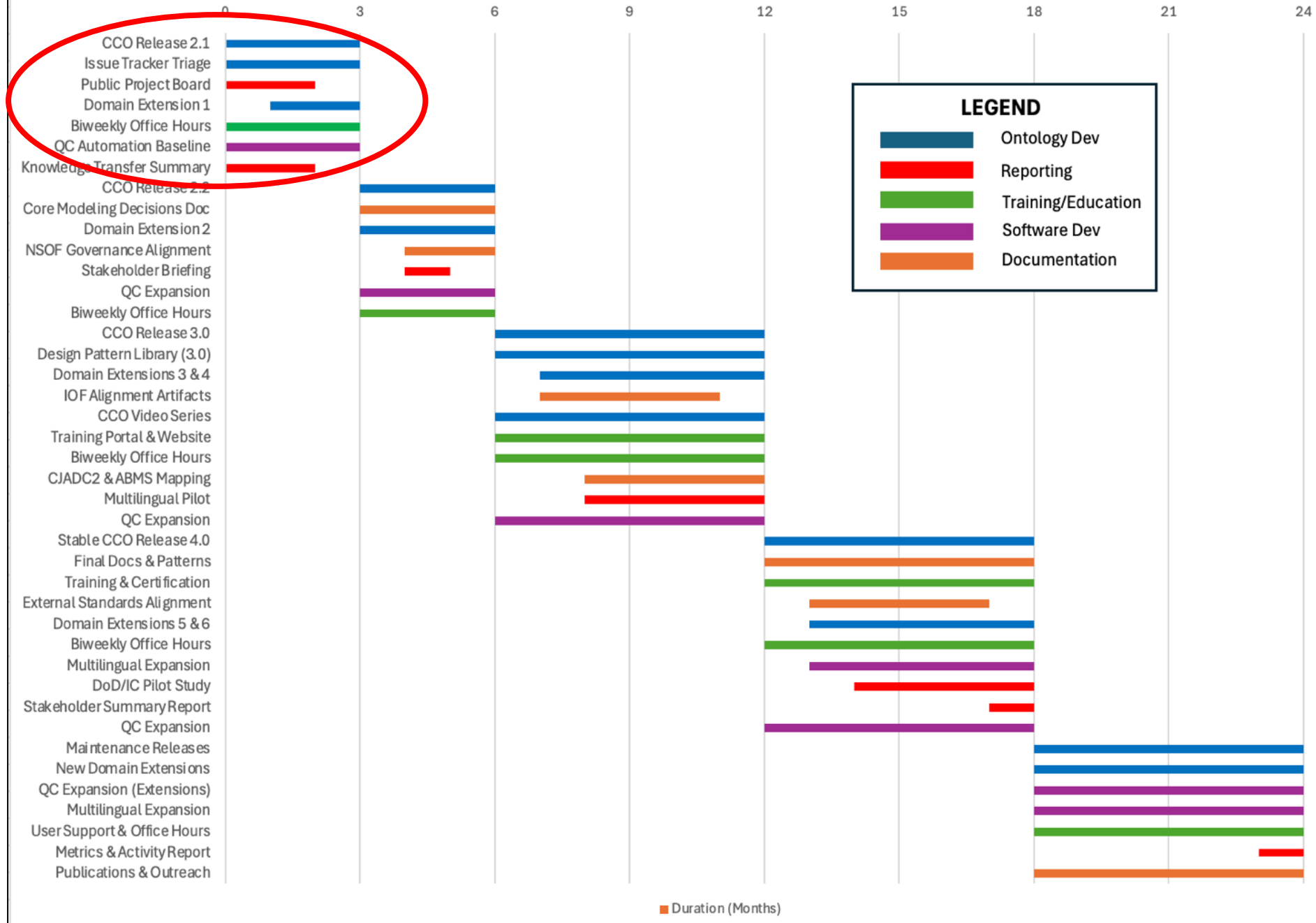
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# CCO Modernization Project



# CCO Modernization Project



| ID   | Deliverable            | Description                                                                                            | Acceptance Criteria                                                         |
|------|------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1A-1 | CCO Release 2.1        | 2.1 release of CCO reflecting prioritized fixes, corrections, and updates identified during transition | Release published with version tag, release notes, and reproducible build   |
| 1A-2 | Issue Tracker Triage   | Systematic review of existing GitHub issues, including resolution, deferral, or closure with rationale | ≥70–80% of legacy issues resolved or dispositioned; labels standardized     |
| 1A-3 | Public Project Board   | GitHub Projects board tracking milestones, issues, and releases                                        | Board publicly accessible and updated weekly                                |
| 1A-4 | Domain Extension 1     | At least one domain-specific extension module scoped, implemented, and integrated with core CCO        | Module versioned, documented, and released alongside CCO 2.1 and QC-covered |
| 1A-5 | Biweekly Office Hours  | Regular stakeholder support sessions addressing modeling, usage, and transition questions              | Sessions held; questions documented and summarized, hosted on website       |
| 1A-6 | QC Automation Baseline | Initial CI/CD pipeline implementing automated ontology quality checks (OWL, SHACL, SPARQL, etc.)       | Automated checks run on PRs and main branch                                 |

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# CCO 2.1

## What's Changed

- Aggregates and Fiat Collections of Processes, Issue 91 by [@mark-jensen](#) in [#113](#)
- Create Readme.md by [@mark-jensen](#) in [#168](#)
- New and changes to political geographical entities by [@mark-jensen](#) in [#171](#)
- Make v1.4 2023-04-07 by [@mark-jensen](#) in [#173](#)
- fixed typo in generically depends on by [@cameronmore](#) in [#198](#)
- Artifact part list typo by [@cameronmore](#) in [#201](#)
- Updating BFO core with latest by [@mark-jensen](#) in [#209](#)
- Build new version 1.5 by [@mark-jensen](#) in [#217](#)
- fixed typo by [@cameronmore](#) in [#274](#)
- improved communication facility definitions by [@cameronmore](#) in [#292](#)
- adjsted definition of gov organization by [@cameronmore](#) in [#289](#)
- Clarified ambiguity in definition by [@cameronmore](#) in [#277](#)
- replaces occurs\_on with occupies temporal region by [@cameronmore](#) in [#275](#)
- clarified priority scale definition by [@cameronmore](#) in [#300](#)
- replaced disposition of aggregate with inheres in aggregate by [@cameronmore](#) in [#279](#)
- Added inhibits from cyber ontology by [@cameronmore](#) in [#268](#)
- 256 use of describes in definitions of velocity and angular velocity by [@cameronmore](#) in [#270](#)
- clarified has object definition by [@cameronmore](#) in [#271](#)
- clarified definitions with no semantic change by [@cameronmore](#) in [#273](#)
- revised loudness definition and added source by [@cameronmore](#) in [#276](#)
- Removed restrictions on definition of mine by [@cameronmore](#) in [#283](#)

## Summary

- Total PR entries: 185
- Unique contributors: 14
- First-time contributors: 7
- Top 3 contributors by PR count: 132 PRs total (70.6% of all listed PRs)
- Top 5 contributors by PR count: 155 PRs total (82.9% of all listed PRs)

| Rank | Contributor                           | PRs | Share |
|------|---------------------------------------|-----|-------|
| 1    | <a href="#">@johnbeve</a>             | 50  | 27.0% |
| 2    | <a href="#">@cameronmore</a>          | 49  | 26.4% |
| 3    | <a href="#">@mark-jensen</a>          | 31  | 16.7% |
| 4    | <a href="#">@oliviahobai</a>          | 12  | 6.4%  |
| 5    | <a href="#">@gregfowlerphd</a>        | 11  | 5.9%  |
| 6    | <a href="#">@neilotte</a>             | 10  | 5.3%  |
| 7    | <a href="#">@shanmukhkalasamudram</a> | 7   | 3.7%  |
| 8    | <a href="#">@APCox</a>                | 5   | 2.7%  |
| 9    | <a href="#">@CarterBeauBenson</a>     | 4   | 2.1%  |
| 10   | <a href="#">@avsculley</a>            | 2   | 1.1%  |
| 11   | <a href="#">@justin2004</a>           | 1   | 0.5%  |
| 12   | <a href="#">@swartik</a>              | 1   | 0.5%  |
| 13   | <a href="#">@oliviamclellan37</a>     | 1   | 0.5%  |
| 14   | <a href="#">@ianknowland</a>          | 1   | 0.5%  |

<https://github.com/CommonCoreOntology/CommonCoreOntologies/releases/tag/v2.1>



develop19 Branches11 TagsGo to fileAdd fileCodeneilotteMerge pull request [#473](#) from gregfowlerphd/463-acts-of-expressi...8ba59d5 · 5 days ago383 Commits.githubMerge pull request [#469](#) from CommonCoreOntology/spa...last weekdocumentationRe-adding the v1.3 CCO docs3 weeks agosrcMerge pull request [#473](#) from gregfowlerphd/463-acts-of...5 days ago.gitignoreUpdate .gitignorelast weekLICENSEUpdate LICENSE2 months agoMakefileso close to a buildlast monthREADME.mdUpdate README.mdlast monthrobots.txtrepo updateslast month

 develop ▾

 19 Branches  11 Tags

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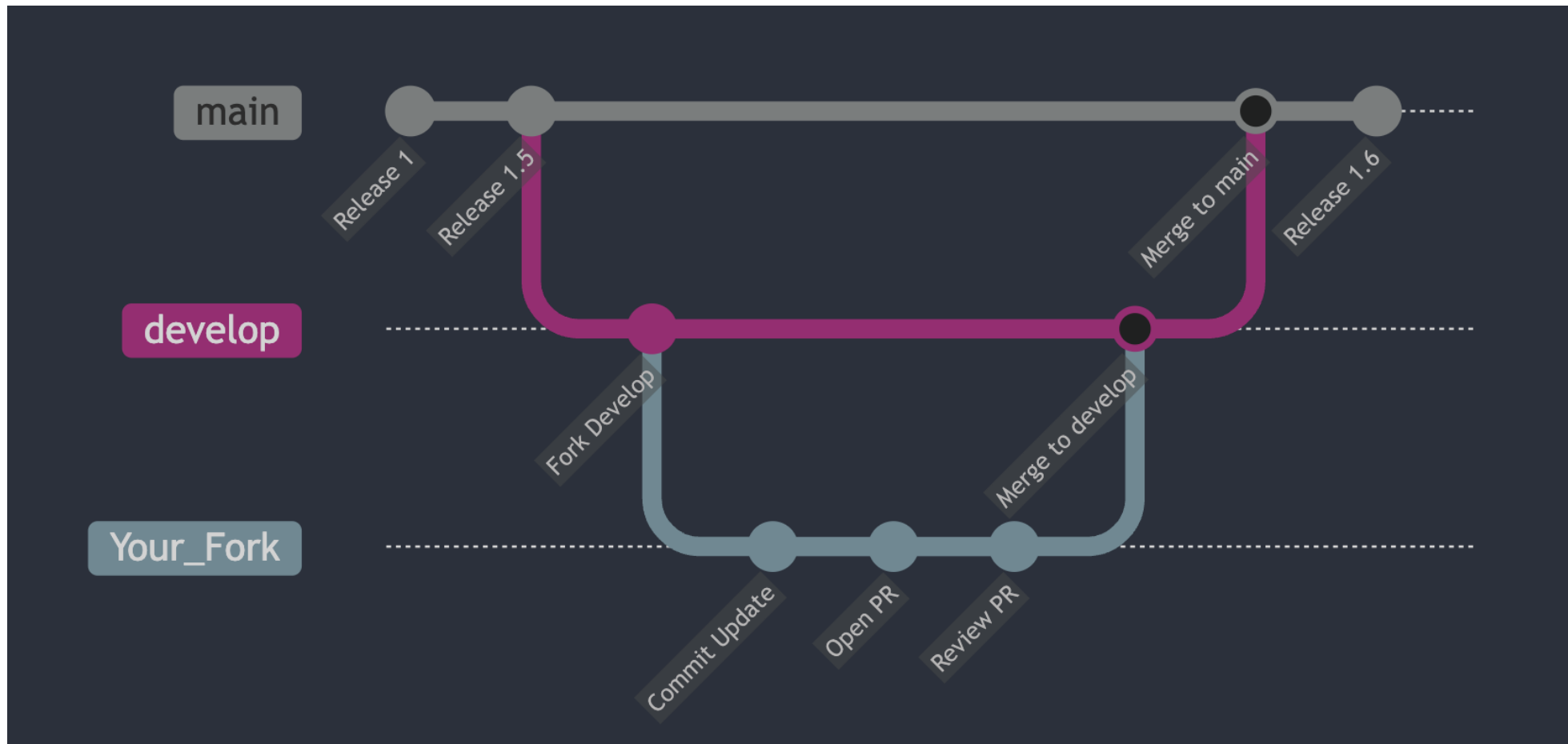
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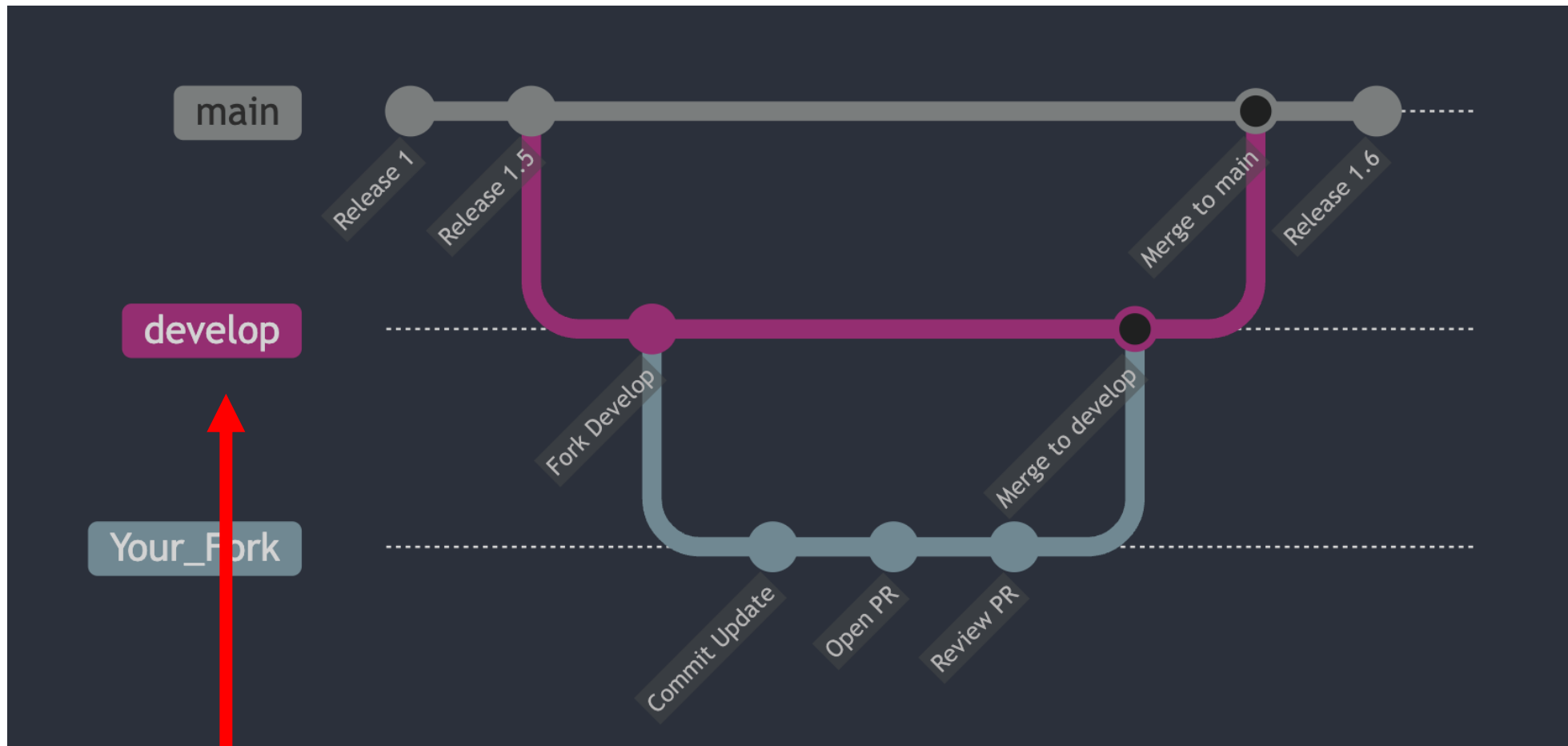
 Code ▾

 neilotte Merge pull request [#473](#) from gregfowlerphd/463-acts-of-expressi...   8ba59d5 · 5 days ago  383 Commits

|                                                                                                 |                                                                           |              |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------|
|  .github       | Merge pull request <a href="#">#469</a> from CommonCoreOntology/spa...    | last week    |
|  documentation | Re-adding the v1.3 CCO docs                                               | 3 weeks ago  |
|  src           | Merge pull request <a href="#">#473</a> from gregfowlerphd/463-acts-of... | 5 days ago   |
|  .gitignore    | Update .gitignore                                                         | last week    |
|  LICENSE      | Update LICENSE                                                            | 2 months ago |
|  Makefile    | so close to a build                                                       | last month   |
|  README.md   | Update README.md                                                          | last month   |
|  robots.txt  | repo updates                                                              | last month   |

develop19 Branches11 TagsGo to fileAdd fileCodeneilotteMerge pull request [#473](#) from gregfowlerphd/463-acts-of-expressi...8ba59d5 · 5 days ago383 Commits.githubMerge pull request [#469](#) from CommonCoreOntology/spa...last weekdocumentationRe-adding the v1.3 CCO docs3 weeks agosrcMerge pull request [#473](#) from gregfowlerphd/463-acts-of...5 days ago.gitignoreUpdate .gitignorelast weekLICENSEUpdate LICENSE2 months agoMakefileso close to a buildlast monthREADME.mdUpdate README.mdlast monthrobots.txtrepo updateslast month



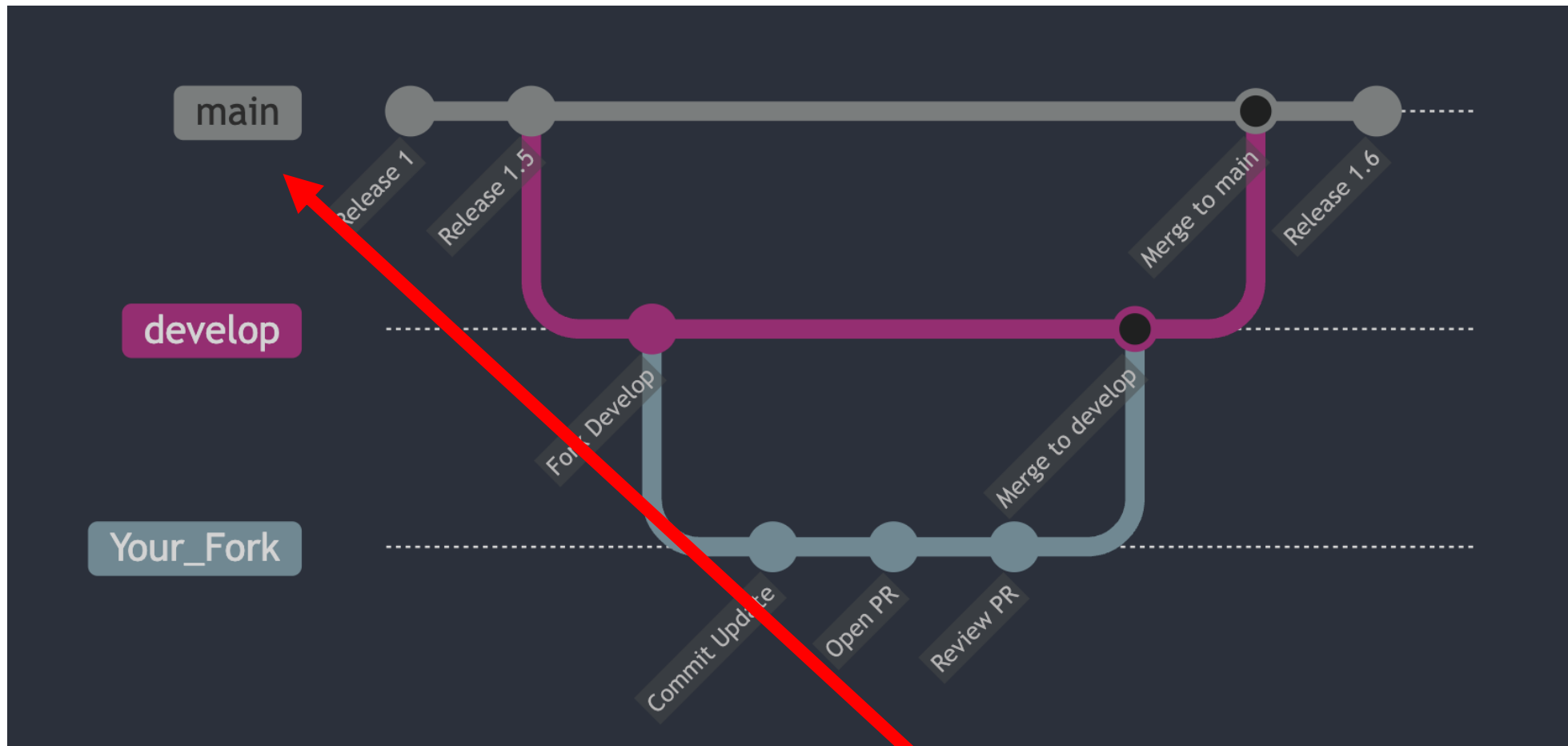


 **CommonCoreOntologies** Public

 **develop** ▾

 **19 Branches**

 **11 Tags**



 **CommonCoreOntologies** Public

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 **develop**  **19 Branches**  **11 Tags**

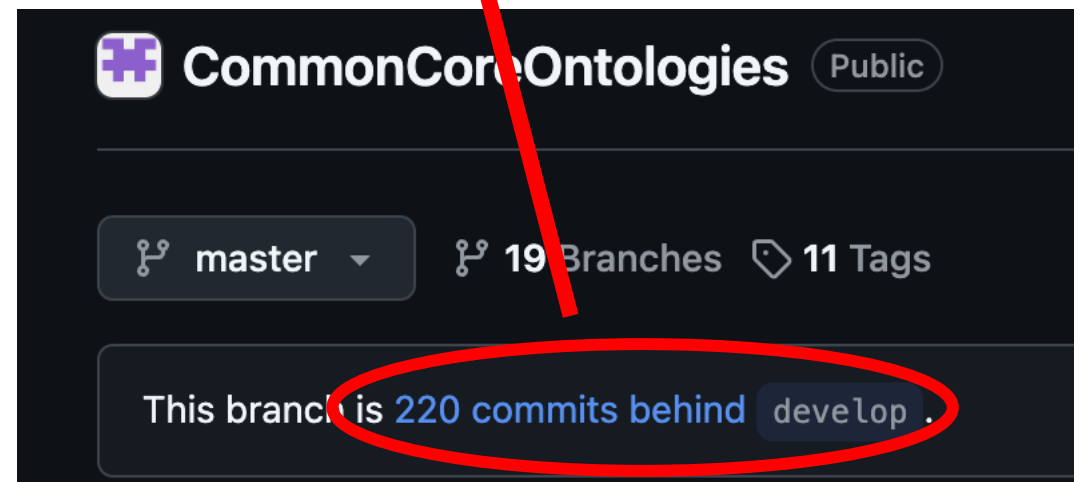
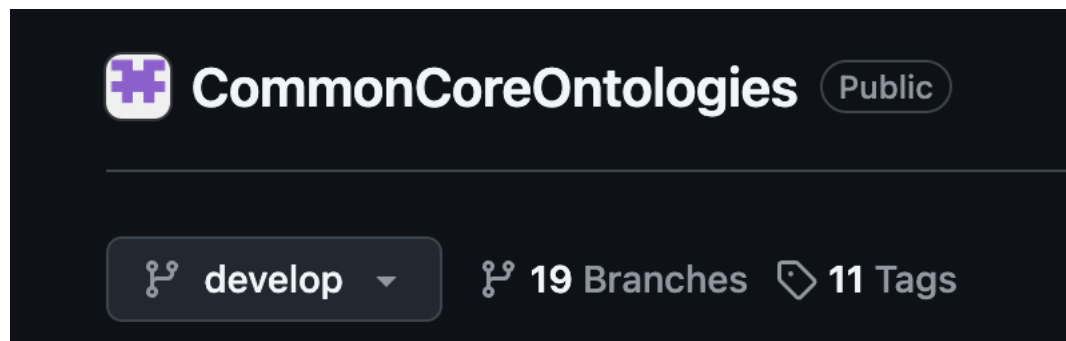
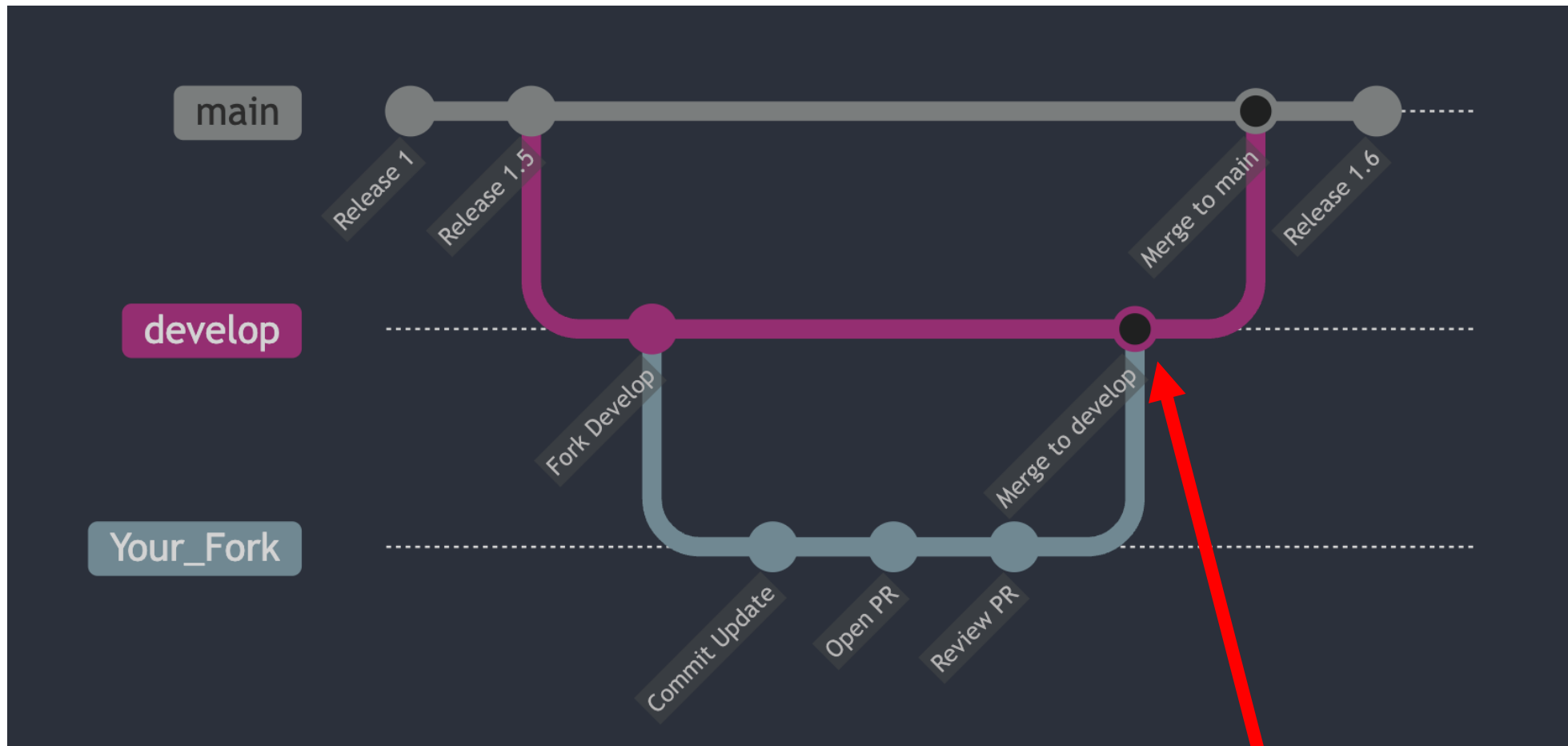
 **CommonCoreOntologies** Public

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 **master**  **19 Branches**  **11 Tags**

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This branch is **220 commits behind** **develop**.



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# *CCO Issue Tracker*

|                          |                                  |                                                                                                                                                           |                                                                                         |                                                                                       |
|--------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <input type="checkbox"/> | <input checked="" type="radio"/> | <b>International Community should be an instance</b> <span>for 2.2 release</span>                                                                         |  11  |    |
|                          |                                  | <b>Bug</b> #855 · APCox opened 3 weeks ago ·  2.2 Release                |                                                                                         |                                                                                       |
| <input type="checkbox"/> | <input checked="" type="radio"/> | <b>Transducer and related terms</b> <span>for 2.2 release</span>                                                                                          |  2   |    |
|                          |                                  | <b>Feature</b> #849 · APCox opened 3 weeks ago ·  2.2 Release            |                                                                                         |                                                                                       |
| <input type="checkbox"/> | <input checked="" type="radio"/> | <b>Definition of Financial Value is Too Restrictive</b> <span>for 2.2 release</span>                                                                      |  13  |    |
|                          |                                  | <b>Bug</b> #766 · seankerm opened on Jan 2 ·  2.2 Release                |                                                                                         |                                                                                       |
| <input type="checkbox"/> | <input checked="" type="radio"/> | <b>Definition of Act of Encounter is problematic</b> <span>for 2.2 release</span>                                                                         |                                                                                         |    |
|                          |                                  | <b>Bug</b> #757 · swartik opened on Oct 29, 2025 ·  2.2 Release          |                                                                                         |                                                                                       |
| <input type="checkbox"/> | <input checked="" type="radio"/> | <b>Act of Military Service should not be a subclass of Act of Employment</b> <span>for 2.2 release</span>                                                 |  3  |   |
|                          |                                  | <b>Feature</b> #756 · swartik opened on Oct 29, 2025 ·  2.2 Release     |                                                                                         |                                                                                       |
| <input type="checkbox"/> | <input checked="" type="radio"/> | <b>The use of 'is boundary of' in several definitions.</b> <span>for 2.2 release</span>                                                                   |  2 |  |
|                          |                                  | <b>Feature</b> #746 · cameronmore opened on Sep 9, 2025 ·  2.2 Release |                                                                                         |                                                                                       |

<https://github.com/CommonCoreOntology/CommonCoreOntologies/issues>

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# CCO Project Board

Backlog

19

Estimate: 0

...

This item hasn't been started

Ready

8

Estimate: 0

...

This is ready to be picked up

In progress

5

Estimate: 0

...

This is actively being worked on

Done

15

Estimate: 0

...

This has been completed

▼

No Priority

47

Estimate: 0

...

CommonCoreOntologies #570

Create a Mapping of CCO to Wikidata

CommonCoreOntologies #527

Create a mapping of CCO to dbpedia

CommonCoreOntologies #184

Spatial and temporal reference systems are directive

CommonCoreOntologies #296

Issue involving Descriptive ICE and several of its subclasses

CommonCoreOntologies #95

Typed data value properties such as 'has integer value'

CommonCoreOntologies #47

Relation between d and c in definitions of "condition described by" and "describes condition"

CommonCoreOntologies #746

The use of 'is boundary of' in several definitions.

CommonCoreOntologies #603

Recording device is a Information processing artifact? Doesn't seem so by the definition.

CommonCoreOntologies #593

Definition of image / represents

CommonCoreOntologies #605

New shortcut property from process to first instant?

CommonCoreOntologies #255

Current version of CCO accessed on web does not import BFO!

CommonCoreOntologies #563

Edit definition of Process Profile

CommonCoreOntologies #444

Artifact Design, Artifact Model, and Artifact Function Specification

CommonCoreOntologies #161

GeographicFeature: broaden definition so as not to require 'natural (i.e. not human made)'

CommonCoreOntologies #861

CCOMerged is missing some new content

CommonCoreOntologies #602

Some Information Bearing Artifact subclass definitions say they bear rather than carry

CommonCoreOntologies #564

Proposal: IBA/IBE/ICE refactor

#674

CommonCoreOntologies #598

Center of Mass shouldn't be a zero-dimensional spatial region

CommonCoreOntologies #776

March 2026 Release Roadmap (PM View)

CommonCoreOntologies #777

Clean up and standardize GitHub labels for CCO

<https://github.com/orgs/CommonCoreOntology/projects/1>

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# Domain Extension

|                                                                                                                                                                |                    |                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotations                                                                                                                                                    | +                  |                                                                                                                                                                             |
| <a href="#">rdfs:label</a>                                                                                                                                     | [language: en-us]  |       |
| Supply Chain Ontology                                                                                                                                          |                    |                                                                                                                                                                             |
| <a href="#">dcterms:license</a>                                                                                                                                | [type: xsd:anyURI] |       |
| <a href="http://opensource.org/licenses/MIT">http://opensource.org/licenses/MIT</a>                                                                            |                    |                                                                                                                                                                             |
| <a href="#">dcterms:abstract</a>                                                                                                                               | [language: en-us]  |       |
| The Supply Chain Ontology is designed to represent generic constructs related to the domain of supply chain and logistics using CCO as its mid-level ontology. |                    |                                                                                                                                                                             |
| <a href="#">dcterms:publisher</a>                                                                                                                              |                    |       |
| Common Core Ontology Governance Board                                                                                                                          |                    |                                                                                                                                                                             |
| <a href="#">dcterms:title</a>                                                                                                                                  | [language: en-us]  |   |
| Common Core Ontologies (CCO) Supply Chain Ontology                                                                                                             |                    |                                                                                                                                                                             |

## Metrics

|                           |     |
|---------------------------|-----|
| Axiom                     | 738 |
| Logical axiom count       | 224 |
| Declaration axioms count  | 128 |
| Class count               | 162 |
| Object property count     | 54  |
| Data property count       | 1   |
| Individual count          | 1   |
| Annotation Property count | 9   |

## Class axioms

|                   |     |
|-------------------|-----|
| SubClassOf        | 131 |
| EquivalentClasses | 43  |
| DisjointClasses   | 0   |
| GCI count         | 7   |
| Hidden GCI Count  | 22  |

# *Domain Extension(s)*

|                                                                                                                      |                                                           |             |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------|
|  BarcodeOntology.ttl                | Fixed stray 'directive' labels to prescriptive (#841)     | last month  |
|  FamilialRelationsOntology.ttl      | extracted barcode ontology, tidy familial ontology (#825) | last month  |
|  ModalRelationOntology.ttl          | Update ModalRelationOntology.ttl                          | 2 years ago |
|  ModalRelationOntologyAdditions.ttl | adding MRO and release Suite (#847)                       | 3 weeks ago |
|  README.md                          | minor tweaks                                              | 2 years ago |
|  catalog-v001.xml                   | Familial relations removal (#814)                         | last month  |

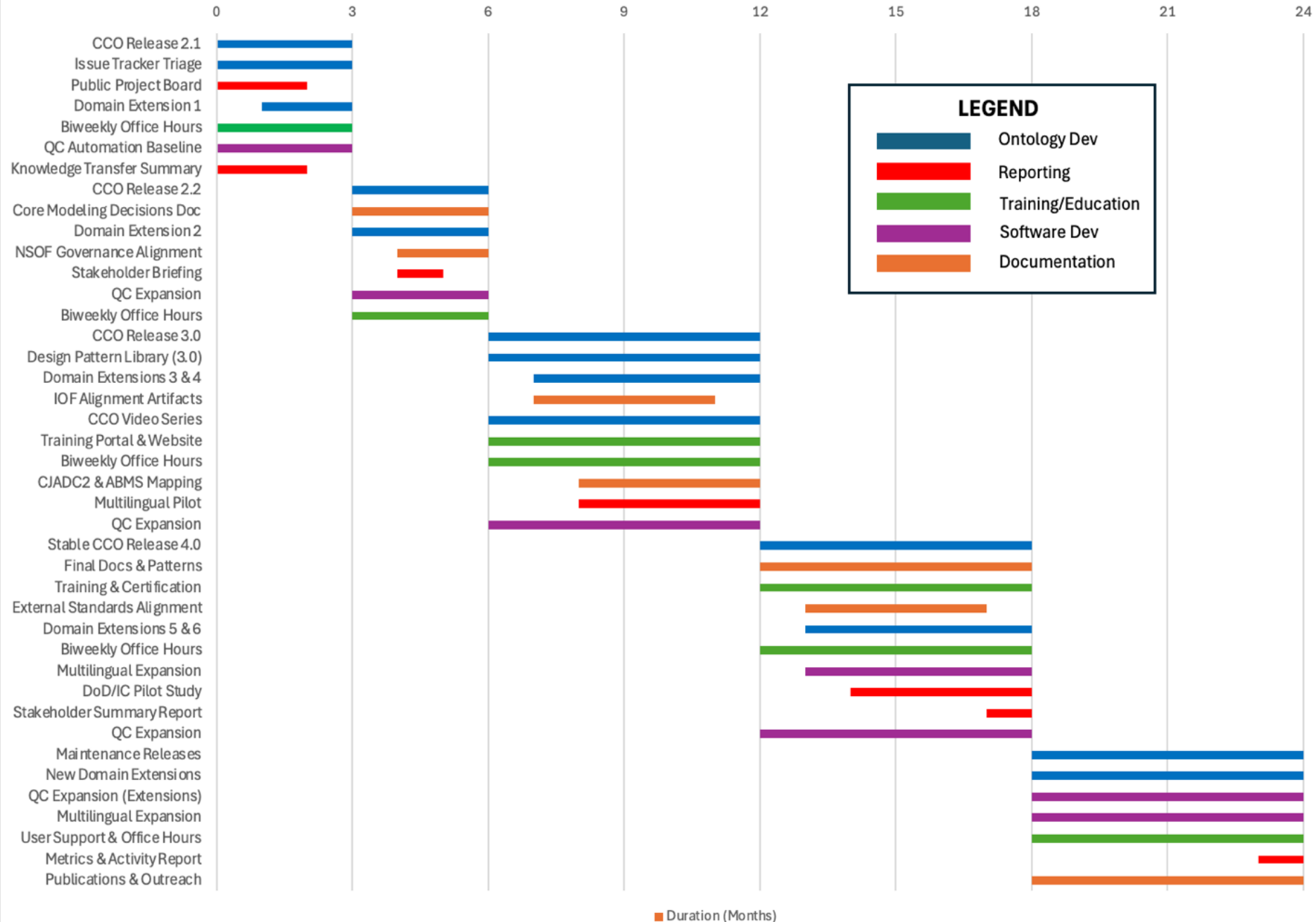
<https://github.com/CommonCoreOntology/CommonCoreOntologies/tree/develop/src/cco-extensions>

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| ID   | Deliverable            | Description                                                                                            | Acceptance Criteria                                                         |
|------|------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1A-1 | CCO Release 2.1        | 2.1 release of CCO reflecting prioritized fixes, corrections, and updates identified during transition | Release published with version tag, release notes, and reproducible build   |
| 1A-2 | Issue Tracker Triage   | Systematic review of existing GitHub issues, including resolution, deferral, or closure with rationale | ≥70–80% of legacy issues resolved or dispositioned; labels standardized     |
| 1A-3 | Public Project Board   | GitHub Projects board tracking milestones, issues, and releases                                        | Board publicly accessible and updated weekly                                |
| 1A-4 | Domain Extension 1     | At least one domain-specific extension module scoped, implemented, and integrated with core CCO        | Module versioned, documented, and released alongside CCO 2.1 and QC-covered |
| 1A-5 | Biweekly Office Hours  | Regular stakeholder support sessions addressing modeling, usage, and transition questions              | Sessions held; questions documented and summarized, hosted on website       |
| 1A-6 | QC Automation Baseline | Initial CI/CD pipeline implementing automated ontology quality checks (OWL, SHACL, SPARQL, etc.)       | Automated checks run on PRs and main branch                                 |



# CCO Modernization Project



# *Outline*

- CCO Governance Board & NCOR
- CCO Modernization Effort
- CCO Future Release Strategy

# CCO Modernization Project



# CCO Modernization Project



| ID   | Deliverable                              | Description                                                                                                  | Acceptance Criteria                                                            |
|------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1B-1 | CCO Release 2.2                          | Refined release incorporating feedback from Phase 1A and remaining transition fixes                          | Release tagged; change log finalized                                           |
| 1B-2 | Revised Core Modeling Decisions Document | Explicit documentation of key modeling choices and constraints, e.g. design patterns and explanations        | Published in repo and website                                                  |
| 1B-3 | Domain Extension 2                       | Second domain module generated; maintenance for domain extension 1                                           | Versioned, documented, and QC-covered and QC-covered for domain extensions 1-2 |
| 1B-4 | NSOF Governance Alignment Plan           | Document describing how CCO aligns with and can inform NSOF governance, releases, and contribution processes | Approved internally and shared with NSOF                                       |
| 1B-5 | QC Expansion                             | Expanded automated checks covering CCO and modules                                                           | ≥75% coverage of CCO 2.2 and module content, with QCs running in CI            |
| 1B-6 | Biweekly Office Hours                    | Regular stakeholder support sessions addressing modeling, usage, and transition questions                    | Sessions held; questions documented and summarized, hosted on website          |

| ID   | Deliverable                              | Description                                                                                                  | Acceptance Criteria                                                            |
|------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1B-1 | CCO Release 2.2<br><b>JUNE 30 2026</b>   | Refined release incorporating feedback from Phase 1A and remaining transition fixes                          | Release tagged; change log finalized                                           |
| 1B-2 | Revised Core Modeling Decisions Document | Explicit documentation of key modeling choices and constraints, e.g. design patterns and explanations        | Published in repo and website                                                  |
| 1B-3 | Domain Extension 2                       | Second domain module generated; maintenance for domain extension 1                                           | Versioned, documented, and QC-covered and QC-covered for domain extensions 1-2 |
| 1B-4 | NSOF Governance Alignment Plan           | Document describing how CCO aligns with and can inform NSOF governance, releases, and contribution processes | Approved internally and shared with NSOF                                       |
| 1B-5 | QC Expansion                             | Expanded automated checks covering CCO and modules                                                           | ≥75% coverage of CCO 2.2 and module content, with QCs running in CI            |
| 1B-6 | Biweekly Office Hours                    | Regular stakeholder support sessions addressing modeling, usage, and transition questions                    | Sessions held; questions documented and summarized, hosted on website          |

# *Future Releases*

- For forthcoming releases NCOR will:
  - Align more closely with the CCO Developer Group
  - Post regular reminders about upcoming changes and dates
  - Approximately one week in advance of a release, create a ‘release branch’ for stakeholder review
  - Aggressively disseminate information about any perceived breaking changes, as well as remediation strategies
  - Publish training and implementation documentation in the repo

| ID   | Deliverable                              | Description                                                                                                  | Acceptance Criteria                                                            |
|------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1B-1 | CCO Release 2.2                          | Refined release incorporating feedback from Phase 1A and remaining transition fixes                          | Release tagged; change log finalized                                           |
| 1B-2 | Revised Core Modeling Decisions Document | Explicit documentation of key modeling choices and constraints, e.g. design patterns and explanations        | Published in repo and website                                                  |
| 1B-3 | Domain Extension 2                       | Second domain module generated; maintenance for domain extension 1                                           | Versioned, documented, and QC-covered and QC-covered for domain extensions 1-2 |
| 1B-4 | NSOF Governance Alignment Plan           | Document describing how CCO aligns with and can inform NSOF governance, releases, and contribution processes | Approved internally and shared with NSOF                                       |
| 1B-5 | QC Expansion                             | Expanded automated checks covering CCO and modules                                                           | ≥75% coverage of CCO 2.2 and module content, with QCs running in CI            |
| 1B-6 | Biweekly Office Hours                    | Regular stakeholder support sessions addressing modeling, usage, and transition questions                    | Sessions held; questions documented and summarized, hosted on website          |



# The Common Core Ontologies

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Cameron MORE <sup>b,d</sup>, Alexander P. COX <sup>b,d</sup>, and John BEVERLEY <sup>b,c,e</sup>

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<sup>b</sup>*National Center for Ontological Research*

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<sup>d</sup>*CUBRC, Inc.*

<sup>e</sup>*Institute for Artificial Intelligence and Data Science*

**Abstract.** The Common Core Ontologies (CCO) are designed as a mid-level ontology suite that extends the Basic Formal Ontology. In 2017, CUBRC, Inc. made CCO openly available. CCO has since been increasingly adopted by a broad group of users and applications and is proposed as the first standard mid-level ontology. Despite these successes, documentation of the contents and design patterns of the CCO has been comparatively minimal. This paper is a step toward providing enhanced documentation for the mid-level ontology suite through a discussion of the contents of the eleven ontologies that collectively comprise the Common Core Ontology suite.

| ID   | Deliverable                              | Description                                                                                                  | Acceptance Criteria                                                            |
|------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1B-1 | CCO Release 2.2                          | Refined release incorporating feedback from Phase 1A and remaining transition fixes                          | Release tagged; change log finalized                                           |
| 1B-2 | Revised Core Modeling Decisions Document | Explicit documentation of key modeling choices and constraints, e.g. design patterns and explanations        | Published in repo and website                                                  |
| 1B-3 | Domain Extension 2                       | Second domain module generated; maintenance for domain extension 1                                           | Versioned, documented, and QC-covered and QC-covered for domain extensions 1-2 |
| 1B-4 | NSOF Governance Alignment Plan           | Document describing how CCO aligns with and can inform NSOF governance, releases, and contribution processes | Approved internally and shared with NSOF                                       |
| 1B-5 | QC Expansion                             | Expanded automated checks covering CCO and modules                                                           | ≥75% coverage of CCO 2.2 and module content, with QCs running in CI            |
| 1B-6 | Biweekly Office Hours                    | Regular stakeholder support sessions addressing modeling, usage, and transition questions                    | Sessions held; questions documented and summarized, hosted on website          |

# CCO Modernization Project



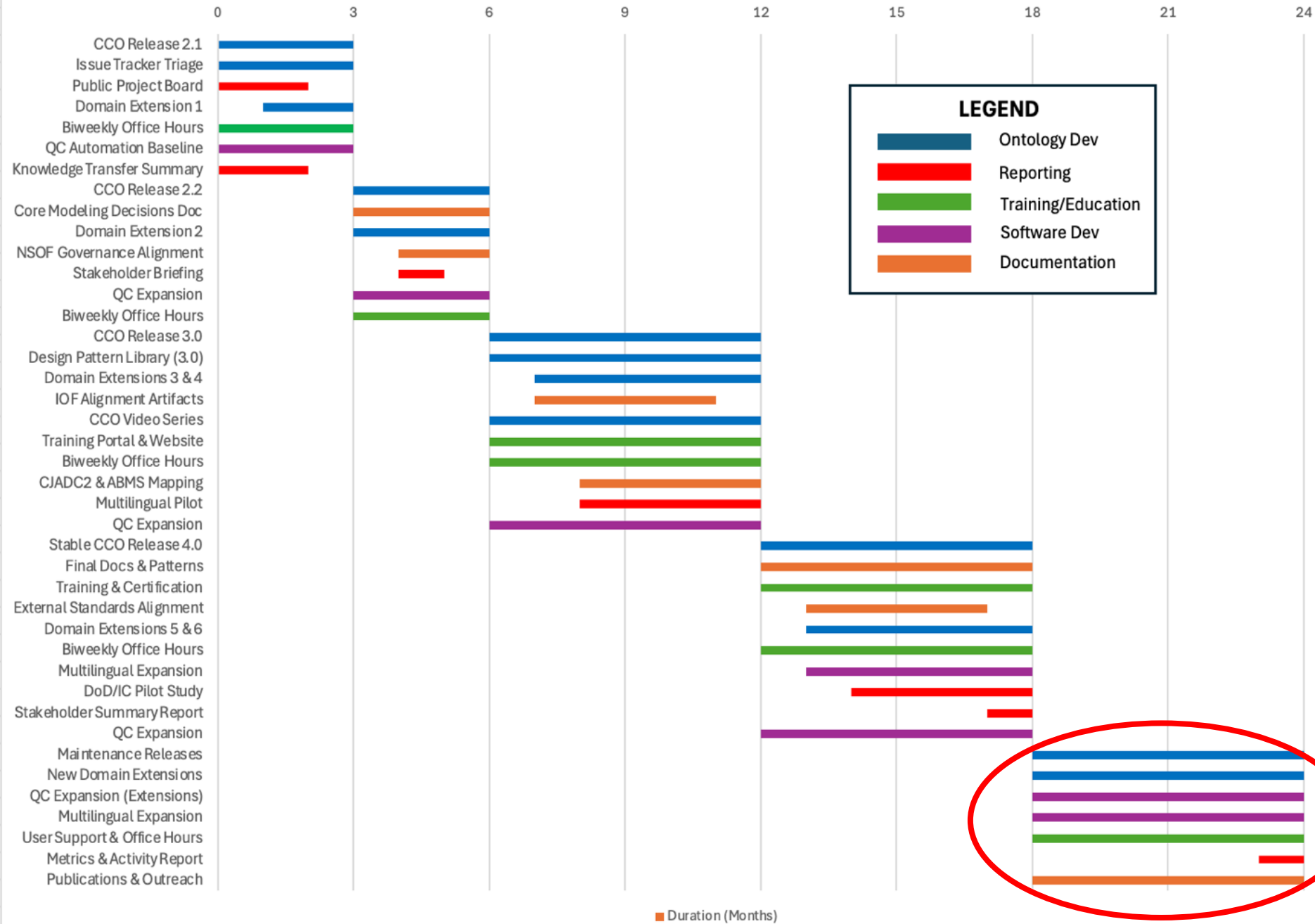
| ID  | Deliverable                           | Description                                                                                     | Acceptance Criteria                                                                                    |
|-----|---------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 2-1 | CCO Release 3.0<br><b>DEC 31 2026</b> | First stable release reflecting refined modeling, patterns, and governance                      | Versioned release with documentation; full formal axiomatizations in OWL, JSON-LD, and CLIF formats    |
| 2-2 | CCO 3.0 Design Pattern Library        | Updated, curated design patterns aligned with CCO 3.0                                           | Published in repo and website and referenced in docs                                                   |
| 2-3 | Domain Extension 3 & 4                | Third and fourth domain module generated; maintenance for domain extensions 1 and 2             | Versioned, documented, and QC-covered and QC-covered for domain extensions 1-4                         |
| 2-4 | IOF Alignment Artifacts               | Mapping or alignment documentation between CCO and Industrial Ontology Foundry artifacts        | Publicly available mapping artifacts                                                                   |
| 2-5 | CCO Video Series                      | Instructional videos covering CCO fundamentals and updates                                      | At least 5 videos published on CCO YouTube channel, captioned                                          |
| 2-6 | Updated CCO Website & Documentation   | Existing website updated to training hub portal, access to lectures, quizzes, FAQs, etc.        | Website live and maintained                                                                            |
| 2-9 | Multilingual Support                  | Develop a proof-of-concept for multilingual support, focusing on one or two priority languages. | CCO 3.0 overlay file with priority language definitions, labels, alternative labels, and documentation |

# CCO Modernization Project



| ID  | Deliverable                                   | Description                                                                                               | Acceptance Criteria                                                                                    |
|-----|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 3-1 | Stable CCO Release 4.0<br><b>JUNE 30 2027</b> | Stabilized low-volatility release                                                                         | Release designated as stable                                                                           |
| 3-2 | Complete Design Pattern & Documentation Suite | Finalized modeling guidance for stabilized CCO 4.0                                                        | Docs complete and versioned                                                                            |
| 3-3 | Training & Certification Materials            | Videos, certification, quizzes, FAQs, and learning materials reflecting CCO 4.0; documented user feedback | Training portal operational and evaluated by users                                                     |
| 3-4 | External Standards Alignments                 | Alignment with GeoSPARQL, QUDT, Time, NIEM, etc.                                                          | Alignment docs published                                                                               |
| 3-5 | Domain Extension 5 & 6                        | Fifth and sixth domain module generated; maintenance on domain extensions 1-4                             | Versioned, documented, and QC-covered for domain extensions 1-6                                        |
| 3-7 | Multilingual Support                          | Refine pilot and expand multilingual support, focusing on two additional languages.                       | CCO 4.0 overlay file with relevant language definitions, labels, alternative labels, and documentation |
| 3-8 | DoD/IC Pilot Integration Study                | Pilot demonstrating CCO use in ISR or AI/ML pipelines                                                     | Study report delivered                                                                                 |
| 3-9 | Stakeholder Summary & Forward Plan            | Report summarizing stabilization and future roadmap                                                       | Delivered to sponsors                                                                                  |

# CCO Modernization Project



| ID  | Deliverable                 | Description                                                                                   | Acceptance Criteria                                                              |
|-----|-----------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| N-1 | Maintenance Release(s)      | Minor updates, fixes, and improvements                                                        | Releases published                                                               |
| N-2 | Domain Extension N          | New domain modules generated                                                                  | Versioned, documented, and QC-covered                                            |
| N-3 | Expanded QC Coverage        | QC automation extended to new domains                                                         | CI coverage increased across new and updated domain module extensions of CCO 4.0 |
| N-4 | Multilingual Expansion      | Additional languages supported                                                                | New language annotations stored in respective overlay files based on CCO 4.0     |
| N-5 | User Support & Office Hours | Continued engagement and support                                                              | SLAs met (responses within 72 hours)                                             |
| N-6 | Metrics & Activity Report   | Usage, adoption, and repo analytics                                                           | Report delivered                                                                 |
| N-8 | Publications                | Continued dissemination of CCO 4.0 and domain extension features, applications, and use cases | Journal publications, proceedings, white papers, presentations, briefs           |



# *What to Tell Your Sponsors*

- Cleanup of artifacts and documentation underway, will be completed **June 30, 2026**
- First wave of major structural changes (3.0) to be released by **December 31, 2026**; will incorporate GeoSPARQL, QUDT, refactoring of information, etc.
- Last wave of major structural changes (4.0) to be released **June 30, 2027**

**Recommendation is to update following 4.0 release**

# *What to Tell Your Sponsors*

- This is at present an ACC-funded effort and as a consequence, AF use cases for domain extensions are prioritized
- Agencies with use cases they would like to see represented are welcome to provide and NCOR will make efforts to accommodate
- Agencies that would like to see their use cases prioritized will need to contribute resources (e.g. people, funds, etc.) to the modernization project